US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250280607
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	KO; Han Seok

SEMICONDUCTOR DEVICE INCLUDING DTI REGION AND METHOD OF MANUFACTURING SAME

Abstract

Proposed are a semiconductor device including a DTI region and a method of manufacturing the same, which enable easy formation of a DTI region including an upper region and a lower region by having an outer surface of the upper region in the DTI region be inclined as the outer surface of the upper region extends downward. A semiconductor device including a DTI region may comprise: a substrate; and a DTI region disposed within the substrate, wherein the DTI region comprises: an upper region extending downward from a surface of the substrate to a first predetermined depth; and a lower region extending downward from a bottom of the upper region to a second predetermined depth within the substrate, wherein the upper region comprises an inclined surface that becomes steeper as an outer surface of the upper region extends downward.

Inventors: **KO; Han Seok (Eumseong-gun, KR)**

Applicant: **DB HiTek Co., Ltd.** (Bucheon-si, KR)

Family ID: **96880787**

Appl. No.: **18/635646**

Filed: **April 15, 2024**

Foreign Application Priority Data

KR	10-2024-0030667	Mar. 04, 2024

Publication Classification

Int. Cl.: **H01L27/02** (20060101); **H01L21/02** (20060101); **H01L21/762** (20060101); **H01L21/764** (20060101)

U.S. Cl.:

CPC **H10D89/10** (20250101); **H01L21/02362** (20130101); **H01L21/76232** (20130101); **H01L21/764** (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to Korean Patent Application No. 10-2024-0030667, filed Mar. 4, 2024, the entire contents of which is incorporated herein for all purposes by this reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present disclosure relates to a semiconductor device including a DTI region and a method of manufacturing the same and, more particularly, to a semiconductor device including a DTI region and a method of manufacturing the same, which enable easy formation of a DTI region including the upper region and a lower region by having an outer surface of the upper region in the DTI region be inclined as the upper region extends downward.

Description of the Related Art

[0003] In recent Bipolar-CMOS-DMOS (BCD) processes, it is required to achieve a high breakdown voltage of 100V or more, and according to this high voltage requirement, a process of forming a deep trench isolation (DTI) region is used to prevent an increase in leakage current through electrical isolation between adjacent devices.

[0004] FIG. **1** is a cross-sectional view showing a semiconductor device including a conventional DTI region.

[0005] Hereinafter, the structure of a conventional DTI region and resulting problems will be described in detail.

[0006] Referring to FIG. **1**, in a conventional semiconductor device **9**, a DTI region **910** may be formed in a substrate **901**. The DTI region **910** is an area gap-filled by an insulating film, and may include: an upper region **911**; and a lower region **913** extending downward from the lower end of the upper region **911** within the substrate **901**. In addition, a pre-metal dielectric (PMD) layer **920** may be formed on the substrate **901**. In the conventional DTI region **910**, the upper region **911** extends downwardly within the PMD layer **920** and the substrate **901** (or STI region **930**) such that the outer surface thereof is substantially perpendicular to the surface of the substrate **901**, while the lower region **913** may be formed to have a tapered width at the lower end of the upper region **911** and extend downward to a predetermined depth.

[0007] In order to form the conventional DTI region **910** as above, after forming a first trench (not shown) in which the upper region **911** will be formed in the PMD layer **920** and the substrate **901**, a second trench (not shown) should be formed below the first trench in the substrate **901**. Then, a process of gap filling the insulating film in the first trench and the second trench is performed. At this time, since the outer surface of the first trench corresponding to the shape of the upper region **911** extends downward so as to be substantially perpendicular to the surface of the substrate **901**, overhang is generated by the gap-filled insulating film. Correspondingly, an etch-back process is performed on the insulating film to lower (reduce) the height (or depth) of the upper part of an air gap AG formed at a relatively high position within the DTI region **910** due to the overhang. Then, the insulating film gap fill process in the first and second trenches is performed again. That is, when forming the conventional DTI region **910**, the overall process may become complicated because the first insulating film gap fill process, the etch-back process, and the second insulating film gap fill process need to be performed after forming the PMD layer **920** on the substrate **901**.

[0008] Moreover, because the upper end of the air gap AG is formed at a relatively high position within the DTI region **910**, the possibility of cracks occurring in the air gap AG cannot be ruled out in a subsequent process. That is, since stress is concentrated in the air gap AG in the DTI region **910**, cracks may occur along the upper side of the air gap AG. When a subsequent process such as a contact process is performed, for example, in a state where a crack occurring in the DTI region, tungsten (W) may remain inside the space created by the crack, causing a defect in the device.

[0009] To solve the above problems, the inventor of the present disclosure proposes a novel semiconductor device with improved structure, the details of which will be described later.

DOCUMENTS OF RELATED ART

[0010] (Patent Document 0001) Korean Patent Application Publication No. 10-2003-0000592 “METHOD FOR MANUFACTURING OF SEMICONDUCTOR DEVICE WITH STI/DTI STRUCTURE”

SUMMARY OF THE INVENTION

[0011] The present disclosure has been made to solve the problems of the related art, and an objective of the present disclosure is to provide a semiconductor device including a DTI region and a method of manufacturing the same, which prevent the upper part of an air gap from being formed at a relatively high height/depth due to overhang on the border of an substrate and a first trench when gap filling an insulating film by making an upper region of a DTI region slanted.

[0012] An objective of the present disclosure is to provide a semiconductor device including a DTI region and a method of manufacturing the same, which prevent defects from occurring in the device during a subsequent process by reducing the possibility of a crack extending to the upper part of an insulating film even if the crack occurs in the insulating film at the upper part of an air gap by lowering (reducing) the height/depth of the upper part of the air gap.

[0013] An objective of the present disclosure is to provide a semiconductor device including a DTI region and a method of manufacturing the same, which improve process efficiency by omitting additional etch-back and secondary gap fill processes by completing a DTI region with one gap fill process in first and second trenches.

[0014] An objective of the present disclosure is to provide a semiconductor device including a DTI region and a method of manufacturing the same, which allow the upper surface of a PMD layer to be flattened by forming a compensation film on an insulating film on a substrate.

[0015] An objective of the present disclosure is to provide a semiconductor device including a DTI region and a method of manufacturing the same, which enable easy control of the overall thickness of a PMD layer by forming a capping layer on a compensation film in some cases.

[0016] The present disclosure may be implemented by an embodiment having the following configuration to achieve the above-described objectives.

[0017] According to an embodiment of the present disclosure, there is provided a semiconductor device including a DTI region. The semiconductor device includes: a substrate; and a DTI region disposed within the substrate, wherein the DTI region may include: an upper region extending downward from a surface of the substrate to a first predetermined depth; and a lower region extending downward from a bottom of the upper region to a second predetermined depth within the substrate, wherein the upper region may include an inclined surface that becomes steeper as an outer surface thereof extends downward.

[0018] According to another embodiment of the present disclosure, in the semiconductor device including a DTI region, an upper part of the upper region may have a lateral width wider than a lateral width of a lower part of the upper region.

[0019] According to still another embodiment of the present disclosure, in the semiconductor device including a DTI region, the inclined surface may have an inclination angle within a range of of 30° to 80°.

[0020] According to still another embodiment of the present disclosure, in the semiconductor device including a DTI region, the DTI region may be formed by a one-time insulating film gap fill process.

[0021] According to still another embodiment of the present disclosure, the semiconductor device including a DTI region may further include: an air gap defined in the DTI region.
[0022] According to still another embodiment of the present disclosure, the semiconductor device including a DTI region may further include: an insulating film disposed on the surface of the substrate, wherein the inclined surface of the upper region may have a first inclination angle, wherein the insulating film may include a second inclined surface that sinks downward at a second inclination angle above the air gap, and wherein the second inclination angle may have a different value from the first inclination angle. According to still another embodiment of the present disclosure, in the semiconductor device including a DTI region, the insulating film may be formed substantially simultaneously with the DTI region.
[0023] According to still another embodiment of the present disclosure, the semiconductor device including a DTI region may further include: a compensation film disposed on the insulating film, wherein the compensation film may have a substantially flat upper surface.
[0024] According to still another embodiment of the present disclosure, the semiconductor device including a DTI region may further include: a capping layer configured to control an overall thickness of a PMD layer, the capping layer being disposed on the compensation film.
[0025] According to still another embodiment of the present disclosure, there is provided a semiconductor device including a DTI region. The semiconductor device includes: a substrate; a gate electrode disposed on the substrate; a drain region disposed on a substrate surface side within the substrate; a source region spaced apart from the drain region and disposed on the substrate surface side within the substrate; a body region surrounding the source region; and a DTI region disposed within the substrate, wherein the DTI region may include: an upper region whose outer surface extends downward from a surface of the substrate to a first predetermined depth; and a lower region extending downward from a bottom of the upper region to a second predetermined depth within the substrate.
[0026] According to still another embodiment of the present disclosure, the semiconductor device including a DTI region may further include: an STI region disposed on a side that overlaps the DTI region within the substrate.
[0027] According to still another embodiment of the present disclosure, in the semiconductor device including a DTI region, the upper region may have a wider lateral width than a lateral width of the lower region.
[0028] According to still another embodiment of the present disclosure, in the semiconductor device including a DTI region, an upper part of the upper region may have a lateral width wider than a lateral width of a lower part of the upper region.
[0029] According to still another embodiment of the present disclosure, the semiconductor device including a DTI region may further include: a first buried layer disposed in the substrate; a second buried layer disposed within the substrate and below the first buried layer; a high voltage well region connected to a side of the second buried layer; and a deep well region surrounding the drain region within the substrate and disposed on the high voltage well region.
[0030] According to still another embodiment of the present disclosure, in the semiconductor device including a DTI region, the upper region may be formed using a mask pattern, wherein the mask pattern is formed such that a side of the substrate where the upper region is to be formed may be open, and outer surfaces of the mask pattern facing each other may be inclined downward and a separation distance between the outer surfaces becomes narrower as the outer surfaces extend downward.
[0031] According to an embodiment of the present disclosure, there is provided a method of manufacturing a semiconductor device including a DTI region. The method includes: forming, at a predetermined location in a substrate, a first trench whose inner walls extend and incline downward; forming a second trench by etching the substrate under the first trench; and completing an insulating film and a DTI region by depositing a first insulating film on the substrate and gap-

filling the first insulating film in the first trench and the second trench.
[0032] According to another embodiment of the present disclosure, the method of manufacturing a semiconductor device including a DTI region may further include: completing a compensation film on the insulating film on the substrate.
[0033] According to still another embodiment of the present disclosure, in the method of manufacturing a semiconductor device including a DTI region, the completing a compensation film may include: depositing a second insulating film on the insulating film; and flattening an upper surface of the second insulating film. According to still another embodiment of the present disclosure, the method of manufacturing a semiconductor device including a DTI region may further include: depositing a capping layer on the compensation film.
[0034] According to still another embodiment of the present disclosure, in the method of manufacturing a semiconductor device including a DTI region, the first trench may be formed so that the inner walls thereof facing each other may be adjacent to each other as the inner walls extend downward.
[0035] The present disclosure has the following effects by the above configurations.
[0036] According to the present disclosure, by making an upper region of a DTI region slanted, it is possible to prevent the upper part of an air gap from being formed at a relatively high height/depth due to overhang on the border of an substrate and a first trench when gap filling an insulating film.
[0037] In addition, according to the present disclosure, by lowering (reducing) the height/depth of the upper part of an air gap to reduce the possibility of a crack extending to the upper part of an insulating film even if the crack occurs in the insulating film at the upper part of the air gap, it is possible to prevent defects from occurring in the device during a subsequent process.
[0038] In addition, according to the present disclosure, by completing a DTI region with one gap fill process in first and second trenches, it is possible to improve process efficiency by omitting additional etch-back and secondary gap fill processes.
[0039] In addition, according to the present disclosure, by forming a compensation film on an insulating film on a substrate, the upper surface of a PMD layer can be flattened.
[0040] Furthermore, according to the present disclosure, by forming a capping layer on a compensation film in some cases, the overall thickness of a PMD layer can be easily controlled.
[0041] Meanwhile, it should be added that even if effects are not explicitly mentioned herein, the effects described in the following specification expected by the technical features of the present disclosure and their potential effects are treated as if they were described in the specification of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0042] The above and other objectives, features, and other advantages of the present disclosure will be more clearly understood from the following detailed description when taken in conjunction with the accompanying drawings, in which:
[0043] FIG. **1** is a cross-sectional view showing a semiconductor device including a conventional DTI region;
[0044] FIG. **2** is a cross-sectional view showing a semiconductor device including a DTI region according to an embodiment of the present disclosure;
[0045] FIG. **3** is an enlarged view of a DTI region according to FIG. **2**; and
[0046] FIGS. **4** to **12** are cross-sectional views showing a method of manufacturing a semiconductor device including a DTI region according to an embodiment of the present disclosure.
DETAILED DESCRIPTION OF THE INVENTION

[0047] Hereinafter, embodiments of the present disclosure will be described in more detail with reference to the accompanying drawings. The embodiments of the present disclosure may be modified in various forms, and the scope of the present disclosure should not be construed as being limited to the following embodiments, but should be construed based on the matters described in the claims. In addition, these embodiments are only provided for reference in order to more completely explain the present disclosure to those of ordinary skill in the art.
[0048] As used herein, the singular form may include the plural form unless the context clearly indicates otherwise. In addition, as used herein, “comprise” and/or “comprising” specify the presence of the recited shapes, numbers, steps, operations, members, elements, and/or groups thereof, but do not exclude the presence or addition of one or more other shapes, numbers, steps, operations, members, elements, and/or groups thereof.
[0049] Hereinafter, it should be noted that when one component (or layer) is described as being disposed on another component (or layer), one component may be disposed directly on another component, or another component(s) or layer(s) may be located between the components. In addition, when one component is expressed as being directly disposed on or above another component, no other component(s) are located between the components. Moreover, being located on “top”, “upper”, “lower”, “top”, “bottom” or “one (first) side” or “side” of a component means a relative positional relationship.
[0050] In addition, it should be noted that, where certain embodiments are otherwise feasible, certain process sequences may be performed other than those described below. For example, two processes described in succession may be performed substantially simultaneously or in the reverse order.
[0051] FIG. **2** is a cross-sectional view showing a semiconductor device including a DTI region according to an embodiment of the present disclosure.
[0052] Hereinafter, a semiconductor device **1** including a DTI region according to an embodiment of the present disclosure will be described in detail with reference to the accompanying drawings.
[0053] Referring to FIG. **2**, the present disclosure relates to a semiconductor device **1** including a DTI region and, more particularly, to a semiconductor device **1** including a DTI region that enables easy formation of a DTI region by having the outer surface of an upper region in the DTI region including the upper region and a lower region be inclined as the outer surface of the upper region extends downward.
[0054] In the semiconductor device **1** including a DTI region according to an embodiment of the present disclosure, a substrate **101** may be formed. A well region used as an active region will be formed in the substrate **101**, and the active region may be defined by an STI region **180** as a device isolation layer. In addition, the substrate **101** may be a substrate doped with a first conductivity type, a P-type diffusion region disposed in the substrate, or may include a P-type epitaxial layer epitaxially grown on the substrate. The STI region **180** may be formed by using a shallow trench isolation (STI) process, but there is no particular limitation thereon. In the drawing, the STI region **180** overlaps a DTI region **190**, but this is only an example. It should be noted that the STI region **180** and a DTI region **190** may be formed in physically distinguishable locations within the substrate **101**.
[0055] A first buried layer **111** and a second buried layer **113** may be formed in the substrate **101**. As an example, the first buried layer **111** may be formed on one side above the second buried layer **113**. In addition, a high voltage well region **120** may be formed to be connected to one side of the second buried layer **113**. The high voltage well region **120** is an impurity doped region (HVNWELL) of a second conductivity type and may be formed within the substrate **101** and on the second buried layer **113**. The first buried layer **111** described above may be an impurity doped region of the first conductivity type, and the second buried layer **113** may be an impurity doped region of the second conductivity type. It should be noted that the first buried layer **111** and the high voltage well region **120** are not essential components of the present disclosure and may be

omitted in some cases.

[0056] A deep well region **130** may be formed within the substrate **101** and on the high voltage well region **120**. One side of the deep well region **130** is connected to the high voltage well region **120**, and the deep well region **130** may be a second conductivity type impurity doped region (DNWELL). In some cases, the deep well region **130** may be formed to be directly connected to the second buried layer **113**.

[0057] In the deep well region **130**, a well region **140** (in a pair **141** and **143**) of the second conductivity type, for example, may be formed, and a drain region **151** may be formed in the first well region **141** and a highly doped region **153** may be formed in the second well region **143**. The drain region **151** is a doped region of the second conductivity type and is doped with impurities at a higher concentration than the first well region **141**. The highly doped region **153** is also a doped region of the second conductivity type and may be doped with impurities at a higher concentration than the second well region **143**. The drain region **151** and the highly doped region **153** may be spaced apart from each other by the STI region.

[0058] In addition, the drain region **151** and the highly doped region **153** are preferably formed on the surface of the substrate **101**. The above-mentioned highly doped region **153** may perform the function of a guard ring together with the second well region **143** and the high voltage well region **120** to reduce leakage current and improve a safe operating area (SOA). The drain region **151** may be electrically connected to a drain electrode (not shown), and the first well region **141** surrounding the drain region **151** is a drain extension region and may improve breakdown voltage characteristics of a high-voltage semiconductor device.

[0059] A body region **160** is formed within the substrate **101**. The body region **160** is a highly doped impurity region of the first conductivity type and may be spaced apart from the deep well region **130** or may be formed to contact each other. A source region **163** is formed in the body region **160** and on the surface side of the substrate **101**. The source region **163** is a region doped with a high concentration of impurities of the first conductivity type and may be electrically connected to a source electrode (not shown). In addition, a body contact region **161** may be formed within the body region **160** and on a side adjacent to or in contact with the source region **163**. The body contact region **161** may be a doped region with a high concentration of impurities of the first conductivity type.

[0060] A gate electrode **170** is formed on the substrate **101**. To be specific, within the active region, the gate electrode **170** may be formed between the drain region **151** and the source region **163**. The gate electrode **170** is formed on a channel region, and the channel region may be turned on or off by a gate voltage applied to the gate electrode **170**. The gate electrode **170** may be made of, for example, any one of conductive polysilicon, metal, conductive metal nitride, and combinations thereof, and may be formed by a CVD, PVD, ALD, MOALD, or MOCVD process, etc.

[0061] A gate insulating film **171** is formed between the gate electrode **170** and the surface of the substrate **101**. The gate insulating film **171** may be made of any one of a silicon oxide film, a high-k dielectric film, and a combination thereof. In addition, the gate insulating film **171** may be formed by an ALD, CVP, or PVD process.

[0062] The sidewall of the gate electrode **170** may be covered with a gate spacer **173**, and the gate spacer **173** may be made of any one of an oxide film, a nitride film, and a combination thereof.

[0063] In addition, the DTI region **190** may be formed within the substrate **101** to a predetermined depth of the substrate **101**. As previously described, the DTI region **190** may be formed at a location that overlaps the STI region **180**, or may be formed at a location that is physically separable from the STI region **180**. The DTI region **190** is an area gap-filled by an insulating film, and may include: an upper region **191**; and a lower region **193** extending downward from the lower end of the upper region **191** within the substrate **101**. As an example, when the DTI region **190** overlaps the STI region **180**, the lower end of the upper region **191** may be formed at substantially the same depth within the substrate **101** as the lower end of the STI region **180**, but the scope of the present disclosure is not limited thereto.

[0064] Below, before describing the DTI region **190** according to an embodiment of the present disclosure in detail, the structure of a conventional DTI region and the problems resulting therefrom will be described in detail.

[0065] Referring to FIG. **1**, in a conventional semiconductor device **9**, a DTI region **910** may be formed in a substrate **901**. The DTI region **910** is an area gap-filled by an insulating film, and may include: an upper region **911**; and a lower region **913** extending downward from the lower end of the upper region **911** within the substrate **901**. In addition, a pre-metal dielectric (PMD) layer **920** may be formed on the substrate **901**. In the conventional DTI region **910**, the upper region **911** extends downwardly within the PMD layer **920** and the substrate **901** (or STI region **930**) such that the outer surface thereof is substantially perpendicular to the surface of the substrate **901**, while the lower region **913** may be formed to have a tapered width at the lower end of the upper region **911** and extend downward to a predetermined depth.

[0066] In order to form the conventional DTI region **910** as above, after forming a first trench (not shown) in which the upper region **911** will be formed in the PMD layer **920** and the substrate **901**, a second trench (not shown) should be formed below the first trench in the substrate **901**. Then, a process of gap filling the insulating film in the first trench and the second trench is performed. At this time, since the outer surface of the first trench corresponding to the shape of the upper region **911** extends downward so as to be substantially perpendicular to the surface of the substrate **901**, overhang is generated by the gap-filled insulating film. Correspondingly, an etch-back process is performed on the insulating film to lower (reduce) the height (or depth) of the upper part of an air gap AG formed at a relatively high position within the DTI region **910** due to the overhang. Then, the insulating film gap fill process in the first and second trenches is performed again. That is, when forming the conventional DTI region **910**, the overall process may become complicated because the first insulating film gap fill process, the etch-back process, and the second insulating film gap fill process need to be performed after forming the PMD layer **920** on the substrate **901**.

[0067] Moreover, because the upper end of the air gap AG is formed at a relatively high position within the DTI region **910**, the possibility of cracks occurring in the air gap AG cannot be ruled out in a subsequent process. That is, since stress is concentrated in the air gap AG in the DTI region **910**, cracks may occur along the upper side of the air gap AG. When a subsequent process such as a contact process is performed, for example, in a state where a crack occurring in the DTI region, tungsten (W) may remain inside the space created by the crack, causing a defect in the device.

[0068] FIG. **3** is an enlarged view of a DTI region according to FIG. **2**.

[0069] Referring to FIGS. **2** and **3**, in order to solve the above problems, the DTI region **190** according to an embodiment of the present disclosure may include the upper region **191** and the lower region **193** extending downward from the lower end of the upper region **191** within the substrate **101**. In this case, the upper region **191** may include an inclined surface **191***a* that becomes steeper as the outer surface thereof extends downward. The cross-section of the inclined surface **191***a* may have a substantially straight or curved shape, or may be formed to partially include both straight and curved shapes. Due to the inclined surface **191***a*, the upper region **191** may have a lateral width with the upper end thereof on the surface side of the substrate **101** is wider than the lower end thereof (W**1**>W**2**). In addition, the inclined surface **191***a* may be formed with an inclination angle (hereinafter referred to as “first inclination angle”) θ within the range of about 30° or more and 80° or less, but the scope of the present disclosure is not limited thereto.

[0070] As described above, in the present disclosure, the lateral width becomes narrower as the upper region **191** extends downward, so that when gap filling an insulating film, the upper end of an air gap AG created due to overhang on the boundary side of a first trench T**1** for the substrate **101** and the upper region **191** may be formed at a relatively low height/depth. As such, when the upper end of the air gap AG is formed at low height/depth, even if a crack occurs in an insulating film **195** at the upper end of the air gap AG, for example, the possibility of the crack extending to the upper part of the insulating film **195** may be reduced. Thus, the possibility of causing defects in the device due to tungsten (W) remaining inside the space created by the crack may be significantly reduced when performing subsequent processes such as a contact process.

[0071] Furthermore, in the present disclosure, when forming the DTI region **190**, a separate insulating film etch-back-secondary gap fill process is not performed, and thus process efficiency may be improved. That is, DTI region **190** is completed with only one gap fill process, and therefore, the insulating film **195** deposited on the surface of the substrate **101** to form the DTI region **190** remains as is.

[0072] Referring to FIG. **2**, a compensation film **197** may be formed on the insulating film **195**. The compensation film **197** is also composed of an insulating film and may form a PMD layer together with the insulating film **195**. The compensation film **197** is a film for flattening the upper surface of the PMD layer, details of which will be explained in a method of manufacturing a semiconductor device to be described later.

[0073] A capping layer **199** including an insulating film may be formed on the compensation film **197**. It should be noted that the capping layer **199** is a layer for controlling the overall thickness of the PMD layer, and is not an essential component of the present disclosure. The above-described insulating film **195**, the compensation film **197**, and the capping layer **199** may include, for example, a borophosphosilicate glass (BPSG) film and a tetraethyl orthosilicate (TEOS) film, and may be made of the same material or different materials, and there is no particular limitation thereon.

[0074] FIGS. **4** to **12** are cross-sectional views showing a method of manufacturing a semiconductor device including a DTI region according to an embodiment of the present disclosure.

[0075] Hereinafter, a method of manufacturing a semiconductor device including a DTI region according to an embodiment of the present disclosure will be described in detail with reference to the attached drawings. Below, for convenience of explanation, only the process of forming a DTI region **190** after well regions within a substrate **101** and a gate region including a gate electrode **170** are formed on the substrate **101** will be described in detail.

[0076] First, a first trench T**1** in which an upper region **191** is to be formed is formed at a predetermined location within the substrate **101** (see FIG. **5**). The first trench T**1** may be formed on a side that overlaps an STI region **180** or may be formed on a side physically separated from the STI region **180** within the substrate **101**, and there is no particular limitation thereon. In this case, the outer surface of the first trench T**1** may be formed to be inclined as the outer surface of the first trench T**1** extends downward. For example, the inner surface of the first trench T**1** may have a substantially straight or curved shape, or may be formed to partially include both straight and curved shapes. In addition, the inner surface of the first trench T**1** may have a lateral width with the upper end thereof is wider than the lower end thereof. As an example, a first inclination angle of the inner surface of the first trench T**1** may be in the range of about 30° or more and 80° or less, but the scope of the present disclosure is not limited thereto.

[0077] In order to form the first trench T**1** with the inclined inner surface in this way, referring to FIG. **4**, a mask pattern PR**1** is created on the substrate **101**. The mask pattern PR**1** is created such that the side where the first trench T**1** is to be formed is open, and it is preferable that outer surfaces PR**1***a* facing each other are inclined in a direction where the separation distance between the outer surfaces PR**1***a* becomes narrower as the outer surfaces PR**1***a* extend downward. Referring to FIG. **5**, after creating the mask pattern PR**1**, the first trench T**1** is completed by performing an etching process on the substrate **101**, and the mask pattern PR**1** is removed.

[0078] The described inclined first trench T**1** may be formed by an etching process using the slope of the outer surface PR**1***a* of the mask pattern PR**1** as shown in FIG. **4**, but may be formed by adjusting an etch recipe or any other process. That is, it should be noted that the first trench T**1** according to an embodiment of the present disclosure may be manufactured using any process that allows the DTI region **190** to have an inclined surface **191***a* as shown in FIG. **3**.

[0079] Thereafter, the substrate **101** under the first trench T**1** is etched to form a second trench T**2** (see FIG. **7**). The second trench T**2** is an area where a lower region **193** will be provided, and may be connected to the lower end of the first trench T**1**. At this time, the outer surface of the second trench T**2** may be formed to extend substantially vertically or may be formed to be inclined, and there is no particular limitation thereon.

[0080] In order to form the second trench T**2**, referring to FIG. **6**, a mask pattern PR**2** is created on the substrate **101**. The mask pattern PR**2** is created such that the side where the second trench T**2** is to be formed is open. Thereafter, referring to FIG. **7**, the second trench T**2** is completed by performing an etching process on the substrate **101** under the first trench T**1**, and the mask pattern PR**2** is removed.

[0081] Thereafter, referring to FIG. **8**, a first insulating film I**1** is deposited on the substrate **101**, and the first insulating film I**1** is gap-filled in the first trench T**1** and the second trench T**2**. Through this process, the DTI region **190** is completed, and an air gap AG may be formed within the DTI region **190**. In addition, an insulating film **195** may be formed on the substrate **101**. In this case, the upper end of the insulating film **195** may remain in a non-flattened state.

[0082] Referring to FIG. **9**, at this time, the above-described upper region **191** and the lower region **193** are formed in the DTI region **190**, and the inclined surface **191***a* may be formed in the upper region **191**. As previously described, the first inclination angle θ of the inclined surface **191***a* may be in the range of about 30° or more and 80° or less, but the scope of the present disclosure is not limited thereto. The insulating film **195** immediately above the air gap AG may also include an inclined surface **195***a* on the upper surface thereof. In this case, the inclined surface **195***a* of the insulating film **195** may have a second inclination angle θ**1** within a predetermined angle range with an imaginary line IL parallel to the substrate **101**. The second inclination angle θ**1** of the inclined surface **195***a* of the insulating film **195** may have a different size from the first inclination angle θ of the inclined surface **191***a* of the upper region **191**, but the scope of the present disclosure is not limited thereto.

[0083] Thereafter, referring to FIG. **10**, a second insulating film **12** is deposited on the insulating film **195** to form a compensation film **197**. Thereafter, referring to FIG. **11**, by performing a CMP process, the compensation film **197** with a substantially flat top surface may be completed.

[0084] Finally, referring to FIG. **12**, the total thickness of a PMD layer may be controlled by additionally forming a capping layer **199** on the compensation film **197**, but it should be noted that this process is not an essential step of the present disclosure.

[0085] The above detailed description is illustrative of the present disclosure. In addition, the above description shows and describes preferred embodiments of the present disclosure, and the present disclosure can be used in various other combinations, modifications, and environments. That is, changes or modifications are possible within the scope of the concept of the disclosure disclosed herein, the scope equivalent to the written disclosure, and/or within the scope of skill or knowledge in the art. The above-described embodiment describes the best state for implementing the technical idea of the present disclosure, and various changes required in the specific application field and use of the present disclosure are possible. Accordingly, the detailed description of the present disclosure is not intended to limit the present disclosure to the disclosed embodiments.

Claims

1. A semiconductor device including a DTI region, the semiconductor device comprising: a substrate; and a DTI region disposed within the substrate, wherein the DTI region comprises: an upper region extending downward from a surface of the substrate to a first predetermined depth; and a lower region extending downward from a bottom of the upper region to a second predetermined depth within the substrate, wherein the upper region comprises an inclined surface

that becomes steeper as an outer surface of the upper region extends downward.
2. The semiconductor device of claim 1, wherein an upper part of the upper region has a lateral width wider than a lateral width of a lower part of the upper region.
3. The semiconductor device of claim 1, wherein the inclined surface has an inclination angle within a range of 30° to 80°.
4. The semiconductor device of claim 1, wherein the DTI region is formed by a one-time insulating film gap fill process.
5. The semiconductor device of claim 1, further comprising: an air gap defined in the DTI region.
6. The semiconductor device of claim 5, further comprising: an insulating film disposed on the surface of the substrate, wherein the inclined surface of the upper region has a first inclination angle, wherein the insulating film comprises a second inclined surface that sinks downward at a second inclination angle above the air gap, and wherein the second inclination angle has a different value from the first inclination angle.
7. The semiconductor device of claim 6, wherein the insulating film is formed substantially simultaneously with the DTI region.
8. The semiconductor device of claim 6, further comprising: a compensation film disposed on the insulating film, wherein the compensation film has a substantially flat upper surface.
9. The semiconductor device of claim 8, further comprising: a capping layer configured to control an overall thickness of a PMD layer, the capping layer being disposed on the compensation film.
10. A semiconductor device including a DTI region, the semiconductor device comprising: a substrate; a gate electrode disposed on the substrate; a drain region disposed on a substrate surface side within the substrate; a source region spaced apart from the drain region and disposed on the substrate surface side within the substrate; a body region surrounding the source region; and a DTI region disposed within the substrate, wherein the DTI region comprises: an upper region whose outer surface extends downward from a surface of the substrate to a first predetermined depth; and a lower region extending downward from a bottom of the upper region to a second predetermined depth within the substrate.
11. The semiconductor device of claim 10, further comprising: an STI region disposed on a side that overlaps the DTI region within the substrate.
12. The semiconductor device of claim 10, wherein the upper region has a wider lateral width than a later width of the lower region.
13. The semiconductor device of claim 10, wherein an upper part of the upper region has a lateral width wider than a lateral width of a lower part of the upper region.
14. The semiconductor device of claim 10, further comprising: a first buried layer disposed in the substrate; a second buried layer disposed within the substrate and below the first buried layer; a high voltage well region connected to a side of the second buried layer; and a deep well region surrounding the drain region within the substrate and disposed on the high voltage well region.
15. The semiconductor device of claim 10, wherein the upper region is formed using a mask pattern, wherein the mask pattern is formed such that a side of the substrate where the upper region is to be formed is open, and outer surfaces of the mask pattern facing each other are inclined downward and a separation distance between the outer surfaces becomes narrower as the outer surfaces extend downward.
16. A method of manufacturing a semiconductor device including a DTI region, the method comprising: forming, at a predetermined location in a substrate, a first trench whose inner walls extend and incline downward; forming a second trench by etching the substrate under the first trench; and completing an insulating film and a DTI region by depositing a first insulating film on the substrate and gap-filling the first insulating film in the first trench and the second trench.
17. The method of claim 16, further comprising: completing a compensation film on the insulating film on the substrate.
18. The method of claim 17, wherein the completing of the compensation film comprises:

depositing a second insulating film on the insulating film; and flattening an upper surface of the second insulating film.

19. The method of claim 17, further comprising: depositing a capping layer on the compensation film.

20. The method of claim 16, wherein the first trench is formed so that the inner walls thereof facing each other are adjacent to each other as the inner walls extend downward.